IIT-JAM MATHEMATICS

Previous Year Paper

2009

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IIT-JAM

MATHEMATICS - 2009

Objective Questions: (Q.1-Q.15)

(1.) Let V be the vector space of all 6×6 real matrices over the field $\mathbb{R}$. Then the dimension of the subspace of V consisting of all symmetric matrices is

(a.) 15

(b.) 18

(c.) 21

(d.) 35

(2.) Let T be the ring of all functions from $\mathbb{R}$ to $\mathbb{R}$ under point-wise addition and multiplication.

Let $I = \{f : \mathbb{R} \to \mathbb{R} \mid f$ is a bounded function$\}$,

$J = \{f : \mathbb{R} \to \mathbb{R} \mid f(3) = 0\}$. Then

(a.) J is an ideal of R but I is not an ideal of R

(b.) I is an ideal of R but J is not an ideal of R

(c.) Both I and J are ideal of R

(d.) Neither I nor J is an ideal of R

(3.) Which of the following sequence of functions is uniformly convergent on $(0,1)$?

(a.) x^n

(b.) $\dfrac{n}{nx+1}$

(c.) $\dfrac{x}{nx+1}$

(d.) $\dfrac{1}{nx+1}$

(4.) Let $T:\mathbb{R}^4 \to \mathbb{R}^4$ be a linear transformation satisfying $T^3+3T^2=4I$, where I is the identity transformation. Then the linear transformation $S=T^4+3T^3-4I$ is

(a.) One-one but not onto

(b.) Onto but not one-one

(c.) Invertible

(d.) Non-invertible

(5.) The number of all subgroups of the group $(\mathbb{Z}_{60}, +)$ of integers modulo 60 is

(a.) 2

(b.) 10

(c.) 12

(d.) 60

(6.) Let $a_n = \begin{cases} \dfrac{1}{3^n}, & \text{if } n \text{ is prime} \\ \dfrac{1}{4^n}, & \text{if } n \text{ is not a prime} \end{cases}$

Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is

(a.) 4

(b.) 3

(c.) $\frac{1}{3}$

(d.) $\frac{1}{4}$

(7.) The set of all limit points of the sequence $1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, \frac{1}{16}, \frac{3}{16}, \frac{5}{16}, \frac{7}{16}, \frac{9}{16}, \ldots$ is

(a.) $[0, 1]$

(b.) $(0, 1]$

(c.) The set of all rational numbers in $[0, 1]$

(d.) The set of all rational numbers in $[0, 1]$ of the form $\frac{m}{2^n}$ where m and n are integers.

(8.) Let $F : \mathbb{R} \to \mathbb{R}$ be a continuous function and $a > 0$. Then the integral $\int_0^a \left[\int_0^x F(y)\,dy\right] dx$ equals

(a.) $\int_0^a yF(y)\,dy$

(b.) $\int_0^a (a-y)F(y)\,dy$

(c.) $\int_0^a (y-a)F(y)\,dy$

(d.) $\int_a^0 yF(y)\,dy$

(9.) The set of all positive values of a for which the series $\sum_{n=1}^{\infty}\left(\frac{1}{n}-\tan^{-1}\left(\frac{1}{n}\right)\right)^a$ converges, is

(a.) $\left(0, \frac{1}{3}\right]$

(b.) $\left(0, \frac{1}{3}\right)$

(c.) $\left[\frac{1}{3}, \infty\right)$

(d.) $\left(\frac{1}{3}, \infty\right)$

(10.) Let a be a non-zero real number.

Then $\lim_{x\to a}\frac{1}{x^2-a^2}\int_a^x \sin(t^2)\,dt$ equals

(a.) $\frac{1}{2a}\sin(a^2)$

(b.) $\frac{1}{2a}\cos(a^2)$

(c.) $-\frac{1}{2a}\sin(a^2)$

(d.) $-\dfrac{1}{2a}\cos\left(a^2\right)$

(11.) Let $T(x, y, z) = xy^2 + 2z - x^2z^2$ be the temperature at the point (x, y, z). The unit vector in the direction in which the temperature decreases most rapidly at $(1, 0, -1)$ is

(a.) $-\dfrac{1}{\sqrt{5}}\hat{i} + \dfrac{2}{\sqrt{5}}\hat{k}$

(b.) $\dfrac{1}{\sqrt{5}}\hat{i} - \dfrac{2}{\sqrt{5}}\hat{k}$

(c.) $\dfrac{2}{\sqrt{14}}\hat{i} + \dfrac{3}{\sqrt{14}}\hat{j} + \dfrac{1}{\sqrt{14}}\hat{k}$

(d.) $-\left(\dfrac{2}{\sqrt{14}}\hat{i} + \dfrac{3}{\sqrt{14}}\hat{j} + \dfrac{1}{\sqrt{14}}\hat{k}\right)$

(12.) Consider the differential equation $2\cos\left(y^2\right)dx - xy\sin\left(y^2\right)dy = 0$. Then

(a.) e^x is an integrating factor

(b.) e^{-x} is an integrating factor

(c.) $3x$ is an integrating factor

(d.) x^3 is an integrating factor

(13.) Suppose $\vec{V} = p(x, y)\hat{i} + q(x, y)\hat{j}$ is a continuously differentiable vector field defined in a domain D

in $\mathbb{R}^2$. Which one of the following statements is NOT equivalent to the remaining ones?

(a.) There exists a function $\phi(x, y)$ such that $\frac{\partial \phi}{\partial x} = p(x, y)$ and $\frac{\partial \phi}{\partial y} = q(x, y)$ for all $(x, y) \in D$

(b.) $\frac{\partial q}{\partial x} = \frac{\partial p}{\partial y}$ holds at all points of D

(c.) $\oint_C \vec{V} \cdot d\vec{r} = 0$ for every piecewise smooth closed curve C in D

(d.) The differential $p\,dx + q\,dy$ is exact in D

(14.) Let, $f, g : [-1,\ 1] \to \mathbb{R}$, $f(x) = x^3, g(x) = x^2|x|$. Then

(a.) f and g are linearly independent on $[-1,\ 1]$

(b.) f and g are linearly dependent on $[-1,\ 1]$

(c.) $f(x)g'(x) - f'(x)g(x)$ is NOT identically zero on $[-1,\ 1]$

(d.) There exist continuous function $p(x)$ and $q(x)$ such that f and g satisfy $y'' + py' + qy = 0$ on $[-1,\ 1]$

(15.) The value of c for which there exists a twice differentiable vector field $\vec{F}$ with curl $\vec{F} = 2x\hat{i} - 7y\hat{j} + cz\hat{k}$ is

(a.) 0

(b.) 2

(c.) 5

(d.) 7

Subjective Questions: (Q.16-Q.29)

(16.) Container A contains 100 cc of milk and container B contains 100 cc of water. 5 cc of the liquid in A is transferred to B, the mixture is thoroughly stirred and 5 cc of the mixture in A is transferred back into A. Each such two-way transfer is called a dilution. Let a_n be the percentage of water in container A after n such dilutions, with the understanding that $a_0 = 0$.

(a.) Prove that $a_1 = \dfrac{100}{21}$ and that, in general,

$$a_n = \frac{100}{21} + \frac{19}{21}a_{n-1} \text{ for } n = 1, 2, 3, \ldots$$

(b.) Using (a) prove that $a_n = 50\left[1 - \left(\dfrac{19}{21}\right)^n\right]$ for $n = 1, 2, 3, \ldots$ Find $\lim\limits_{n\to\infty} a_n$ and explain why the answer is intuitively obvious.

(17.) (a.) Let $f:\mathbb{N}\times\mathbb{N}\to\mathbb{R}$ be a non-negative function. Assume that for every $m\in\mathbb{N}$, the series $\sum_{n=1}^{\infty} f(m,n)$ is convergent and has sum a_m and further that the series $\sum_{m=1}^{\infty} a_m$ is also convergent and has sum L. Prove that for every n, the series $\sum_{m=1}^{\infty} f(m,n)$ is convergent and if we denote its sum by b_n then the series $\sum_{n=1}^{\infty} b_n$ is also convergent and has sum L.

(b.) Define $f:\mathbb{N}\times\mathbb{N}\to\mathbb{R}$ by

$$f(m,n)=\begin{cases}0 & \text{if } n>m,\\ \dfrac{-1}{2^{m-n}} & \text{if } n<m,\\ 1 & \text{if } n=m.\end{cases}$$

Show that $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} f(m,n)=2$

and $\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} f(m,n)=0$.

(18.) (a.) Evaluate $\iint_R \cos\left(\max\{x^3, y^{3/2}\}\right)dx\,dy$,

where $R=[0,1]\times[0,1]$.

(b.) Let $S = \sqrt{1} + \sqrt{2} + \sqrt{3} + \ldots + \sqrt{10000}$ and $I = \int_0^{10000} \sqrt{x}\, dx$. Show that $I \le S \le I + 100$.

(19.) Let $D = \{(x, y) : x \ge 0, y \ge 0\}$.

Let $f(x, y) = (x^2 + y^2)e^{-x-y}$ for $(x, y) \in D$.

Prove that f attains its maximum on D at two boundary points. Deduce that $\frac{x^2 + y^2}{4} \le e^{x+y-2}$ for all $x \ge 0, y \ge 0$.

(20.) (a.) Let $a_1, b_1, a_2, b_2 \in \mathbb{R}$. Show that the condition $a_2 b_1 > 0$ is sufficient but not necessary for the system

$$\frac{dx}{dt} = a_1 x + b_1 y,$$

$$\frac{dy}{dt} = a_2 x + b_2 y,$$

to have two linearly independent solutions of the form $x = c_1 e^{\lambda_1 t}$, $y = d_1 e^{\lambda_1 t}$ and $x = c_2 e^{\lambda_2 t}$, $y = d_2 e^{\lambda_2 t}$ with $c_1, d_1, c_2, d_2, \lambda_1, \lambda_2 \in \mathbb{R}$.

(b.) Show that the differential equation representing the family of all straight lines which have an intercept of constant length L between the coordinate axes is

$$x\frac{dy}{dx} - y = \frac{L\frac{dy}{dx}}{\sqrt{1+\left(\frac{dy}{dx}\right)^2}}$$

(21.) Let $A, B, k > 0$. Solve the initial value problem $\frac{dy}{dx} - Ay + By^3 = 0, x > 0, y(0) = k$.

Also show that

(a.) If $k < \sqrt{\frac{A}{B}}$, then the solution $y(x)$ is monotonically increasing on $(0, \infty)$ and tends to $\sqrt{\frac{A}{B}}$ as $x \to \infty$.

(b.) If $k > \sqrt{\frac{A}{B}}$, then the solution $y(x)$ is monotonically decreasing on $(0, \infty)$ and tends to $\sqrt{\frac{A}{B}}$ as $x \to \infty$.

(22.) (a.) Evaluate $\int_C (3y^2 + 2z^2)\,dx + (6x - 10z)\,y\,dy + (4xz - 5y^2)\,dz$ along the portion from $(1, 0, 1)$ to $(3, 4, 5)$ of the curve C, which is the intersection of the two surfaces $z^2 = x^2 + y^2$ and $z = y + 1$.

(b.) A particle moves counterclockwise along the curve $3x^2+y^2=3$ from $(1,\ 0)$ to a point P, under the action of the force $\vec{F}(x,y)=\frac{x}{y}\hat{i}+\frac{y}{x}\hat{j}$. Prove that there are two possible locations of P such that the work done by $\vec{F}$ is 1.

(23.) Verify Stokes's theorem for the hemisphere $x^2+y^2+z^2=9$, $z\geq 0$ and the vector field $\vec{F}=(z^2-y)\hat{i}+(x-2yz)\hat{j}+(2xz-y^2)\hat{k}$.

(24.) (a.) Let $T:\mathbb{R}^3\to\mathbb{R}^2$ be the linear transformation defined by $T(x,y,z)=(x+2y,\ x-z)$.

Let $N(T)$ be the null space of T and

$W=\{\vec{v}\in\mathbb{R}^3 \mid \vec{v}\cdot\vec{u}=0, \text{ for all } \vec{u}\in N(T)\}$.

Find a linear transformation $S:\mathbb{R}^2\to W$ such that $TS=I$, where I is the identity transformation on $\mathbb{R}^2$.

(b.) Suppose A is a real square matrix of odd order such that $A+A^T=0$. Prove that A is singular.

(25.) (a.) Find all pairs $(a,\ b)$ of real numbers for which the system of equations

$$x+3y=1$$

$$4x + ay + z = 0$$

$$2x + 3z = b$$

has (i) a unique solution,

(ii) infinitely many solutions,

(iii) no solution.

(b.) Let A be an $n \times n$ matrix such that $A^n = 0$ and $A^{n-1} \neq 0$. Show that there exists a vector $v \in \mathbb{R}^n$ such that $\{v, Av, \ldots, A^{n-1}v\}$ forms a basis for $\mathbb{R}^n$.

(26.) (a.) In which of the following pairs are the two groups isomorphic to each other? Justify your answers.

(i) $\mathbb{R}/\mathbb{Z}$ and S^1, where $\mathbb{R}$ is the additive group of real numbers and $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ under complex multiplication.

(ii) $(\mathbb{Z}, +)$ and $(\mathbb{Q}, +)$.

(b.) Prove or disprove that if G is a finite abelian of order n, and k is a positive integer which divides n, then G has at most one subgroup of order k.

(27.) Let I and J be ideals of a ring R. Let IJ be the set of all possible sums $\sum_{i=1}^{n} a_i b_i$, where $a_i \in I$, $b_i \in J$ for $i = 1, 2, \ldots, n$ and $n \in \mathbb{N}$.

(a.) Prove that IJ is an ideal of R and $IJ \subseteq I \cap J$.

(b.) Is it true that $IJ = I \cap J$? Justify your answer.

(28.) A sequence $\{f_n\}$ of functions defined on an interval I is said to be uniformly bounded on I if there exists some M such that $|f_n(x)| \le M$ for all $x \in I$ and for all $n \in \mathbb{N}$.

(a.) Prove that if a sequence of function $\{f_n\}$ converges to a function f on I and $\{f_n\}$ is uniformly bounded on I, then f is bounded on I.

(b.) Suppose the sequences $\{f_n\}$ and $\{g_n\}$ of functions converge uniformly to f and g respectively on I and both are uniformly bounded on I. Prove that the product sequence $\{f_n g_n\}$ converges to fg uniformly on I. Show by an example that this may fail if only one of $\{f_n\}$ and $\{g_n\}$ is uniformly bounded on I.

(29.) (a.) Prove that if f is a real-valued function which is uniformly continuous on an interval $(a,\ b)$, then f is bounded on $(a,\ b)$.

(b.) Let f be a differentiable function on an interval (a, b). Assume that f' is bounded on (a, b). Prove that f is uniformly continuous on (a, b).

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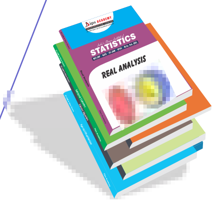

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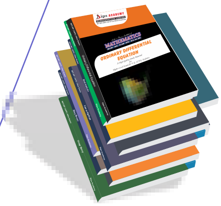

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